



SGALAB

Evolution with the world

SxLAB Family Member

SGALAB Quick Start

(3) Matrix Variables Inputs
and
Applications in Micro and Macro Economics

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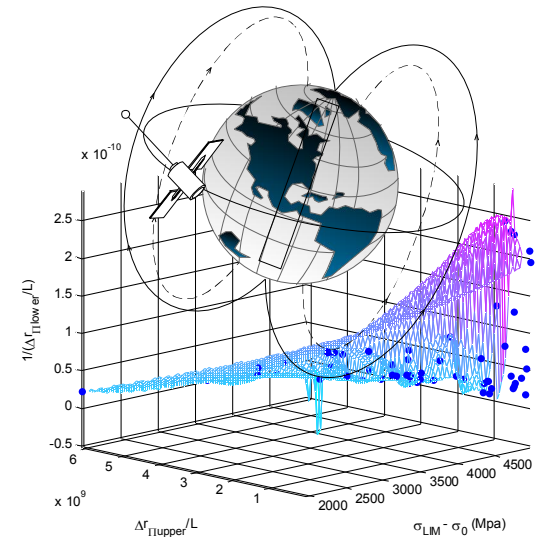
What is matrix variable inputs

Parameter setting and files

Macro Economic case – agricultural system

Micro Economic case – exchange rates determination

FAQ



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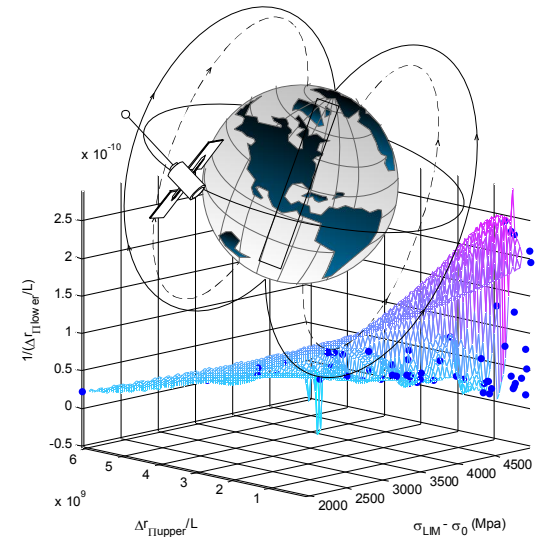
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What is matrix variable inputs

Linear case

$$J = \text{fitness}(X) = \text{Minimise} : |AX - B|$$

where

A is $M \times N$ matrix,

X is $N \times R$ matrix, with the element x_{ij} ,

$$i = 1, 2, \dots, N, j = 1, 2, \dots, R$$

X is the matrix variable for GA

B is $M \times R$ matrix.

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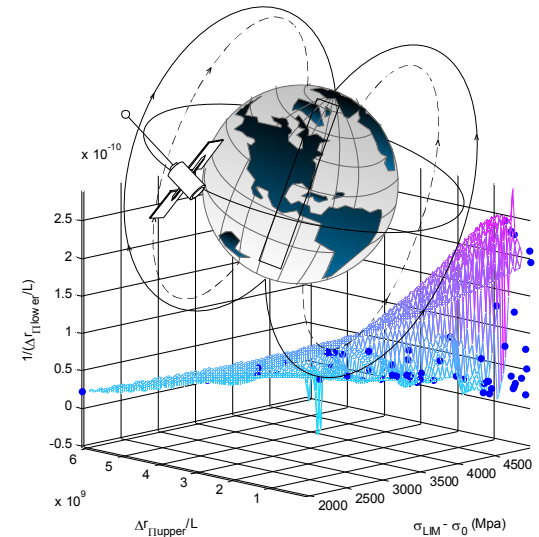
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Parameter setting and files

case 1 – with same dimensions
,the same initials

$$J = \text{fitness}(X, Y, Z)$$



3 matrix variables

where

X, Y, Z have *same* dimension, $M \times N$, e.g. 7×13

all the x_{ij} have *same* initial range, e.g. $[-1, 4]$,

all the y_{ij} have *same* initial range, e.g. $[-10, 100]$,

all the z_{ij} have *same* initial range, e.g. $[1, 30]$,

and $i = 1, 2, \dots, M$,

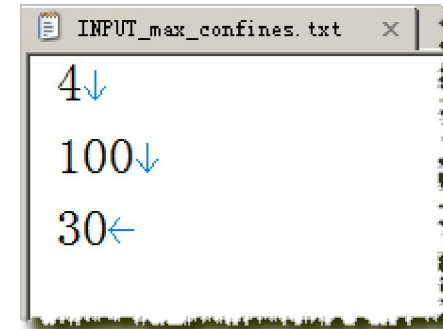
$j = 1, 2, \dots, N$

Parameter setting and files

case 1 – with same dimensions
,the same initials

Run `SGALAB_demo_SO_MATRIX_muGA.m`

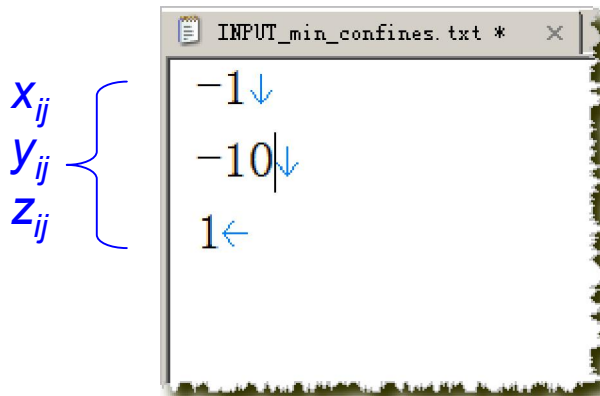
`INPUT_max_confines.txt`



4↓
100↓
30←

x_{ij}
 y_{ij}
 z_{ij}

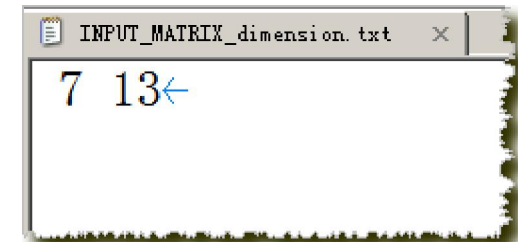
`INPUT_min_confines.txt`



-1↓
-10↓
1←

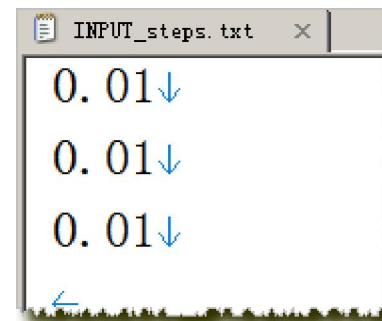
x_{ij}
 y_{ij}
 z_{ij}

`INPUT_MATRIX_dimension.txt`



7 13←

`INPUT_steps.txt`



0.01↓
0.01↓
0.01↓
←

x_{ij}
 y_{ij}
 z_{ij}

`SGA_FITNESS_function.m`



```
function [ fitness ] = SGA_FITNESS_function( X , Y , Z )↓  
↓
```

Parameter setting and files

case 2 – with same dimensions
,the different initials

$$J = \text{fitness}(X, Y, Z)$$



3 matrix variables

where

X, Y, Z have *same* dimension, $M \times N$, e.g. 7×13

all the x_{ij} have *different* initial range, e.g. $[x_{ij0}, x_{ij1}]$,

all the y_{ij} have *different* initial range, e.g. $[y_{ij0}, y_{ij1}]$,

all the z_{ij} have *different* initial range, e.g. $[z_{ij0}, z_{ij1}]$,

and $i = 1, 2, \dots, M$,

$j = 1, 2, \dots, N$



Parameter setting and files

case 3 – with same dimensions
with both the same and
the different initials

$$J = \text{fitness}(X, Y, Z)$$



3 matrix variables

where

X, Y, Z have *same* dimension, $M \times N$, e.g. 7×13

all the x_{ij} have *same* initial range, e.g. $[-1, 4]$,

all the y_{ij} have *different* initial range, e.g. $[y_{ij0}, y_{ij1}]$,

all the z_{ij} have *different* initial range, e.g. $[z_{ij0}, z_{ij1}]$,

and $i = 1, 2, \dots, M$,

$j = 1, 2, \dots, N$



Parameter setting and files

case 4 – with different dimensions
,the same initials

$$J = \text{fitness}(X, Y, Z)$$

 3 matrix variables

where

X, is $MX \times NX$ matrix, e.g. 7×13 ,

Y, is $MY \times NY$ matrix, e.g. 1×3 ,

Z, is $MZ \times NZ$ matrix, e.g. 4×4 ,

all the x_{ij} have *same* initial range, e.g. $[-1, 4]$,

all the y_{ij} have *same* initial range, e.g. $[-10, 100]$,

all the z_{ij} have *same* initial range, e.g. $[1, 30]$,

and $i = 1, 2, \dots, MX$ (MY, MZ),

$j = 1, 2, \dots, NX$ (NY, NZ)



Parameter setting and files

case 5 – with different dimensions
,the different initials

$$J = \text{fitness}(X, Y, Z)$$

 3 matrix variables

where

X, is $MX \times NX$ matrix, e.g. 7×13 ,

Y, is $MY \times NY$ matrix, e.g. 1×3 ,

Z, is $MZ \times NZ$ matrix, e.g. 4×4 ,

all the x_{ij} have *different* initial range, e.g. $[x_{ij0}, x_{ij1}]$,

all the y_{ij} have *different* initial range, e.g. $[y_{ij0}, y_{ij1}]$,

all the z_{ij} have *different* initial range, e.g. $[z_{ij0}, z_{ij1}]$,

and $i = 1, 2, \dots, MX$ (MY, MZ),

$j = 1, 2, \dots, NX$ (NY, NZ)



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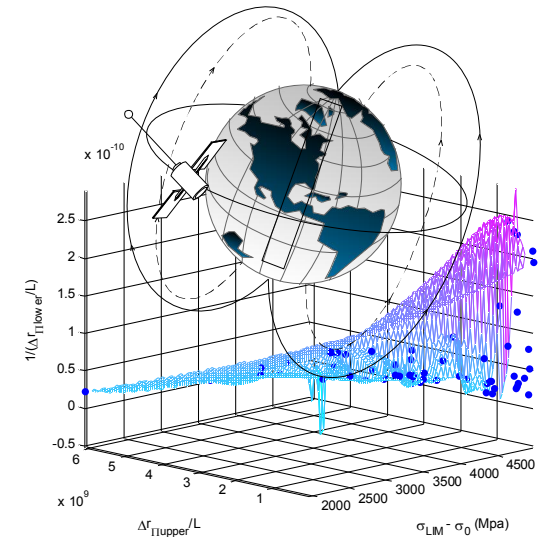
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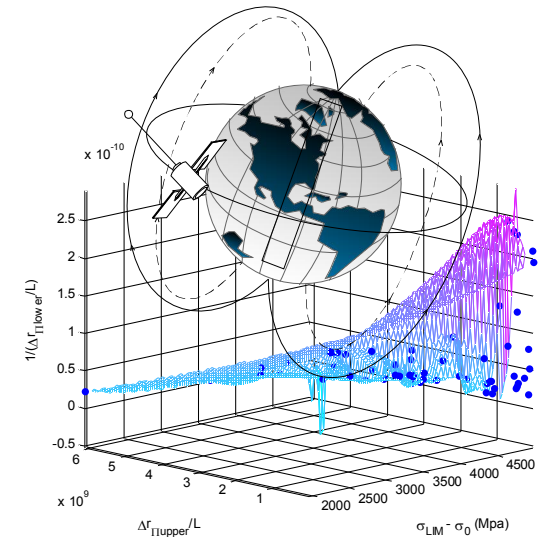
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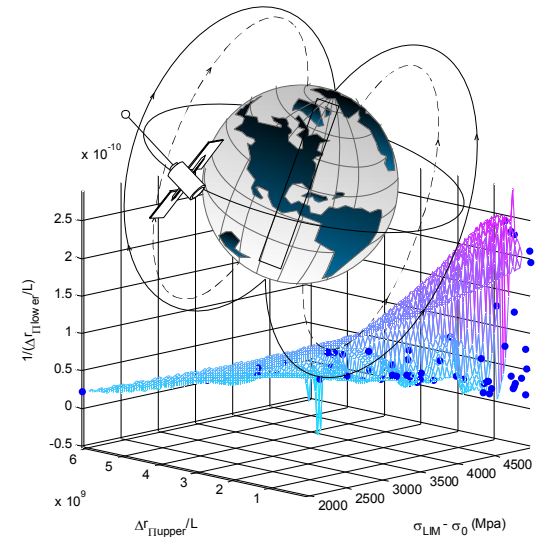
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End

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八声甘州·清秋

遍古今长河水东流，千年风雨稠。
帝尧部族传，夏商祭幻，烽火二周。
秦俑汉相玄奘，瘦金河图上。
古道断西风，史外话万象。

曾经勤学路远，昨夜书万卷，朝田暮殿。
剑影落叶旋，浅酒菊花粥。
烟波一伞秋色满，缘浅灯深窗半悬。
谁如烟，渺渺诗笺，明月正圆。